A project report on

Wave-Driven Rotary Energy Generation System for Coastal Communities

submitted in partial fulfillment of the requirements for the degree of

B. Tech In
Electronics and Computer Science Engineering

By

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CERTIFICATE

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in partial fulfilment of the requirements for the award of the **Degree of Bachelor of Technology** in **ELECTRONICS AND COMPUTER SCIENCE** is a bonafide record of the work carried out under my guidance and supervision at School of Electronics Engineering, KIIT (Deemed to be University).

Signature of Supervisor
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ACKNOWLEDGEMENTS

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ABSTRACT

The growing demand for the use of sustainable and renewable forms of energy has led to the exploration of alternative methods for generating power. The current project proposes the design of a Wave-Driven Rotary Energy Generation System, which can be utilized for the efficient conversion of wave energy into electrical power. The system works on the principle of utilizing the kinetic energy produced due to the motion of the waves, which is then converted into mechanical form using the motor-driven wave generation system. The mechanical form of the wave's motion is then utilized for the compression of air, which in turn drives the turbine connected to the micro-generator, thus generating electrical power.

For the efficient generation of electrical power, the generated power is regulated using rectifiers and capacitors, thus helping in the stabilization of voltage. In addition, the IoT system has also been incorporated using the microcontroller, thus helping in the efficient tracking of voltage, wave intensity, and the overall performance of the system.

The proposed system is efficient, cost-effective, eco-friendly, and can be implemented in coastal areas where conventional sources of energy are scarce. The system also provides the advantage of further development for large-scale implementation. The current project proposes the efficient conversion of wave energy into electrical power using the hybrid approach.

The current system contributes to the development of renewable energy, thus helping in the achievement of the concept of sustainable development through the efficient utilization of the natural resources present in the ocean.

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CHAPTER 1

INTRODUCTION

1.1 Background Studies /Literature Survey

In the recent past, the need for renewable forms of energy has increased tremendously due to the exhaustion of fossil fuels and the growing concern over environmental pollution. Among the different forms of renewable energy available, wave energy has received much attention in recent years because of its high energy density and predictability. Unlike solar and wind energy, wave energy is relatively unaffected by weather conditions and provides a constant output of power.

The wave energy conversion devices are capable of harnessing the kinetic and potential energy of the waves and converting it into a more useful form of energy. Several techniques have been proposed for wave energy conversion devices. These techniques include oscillating water columns, point absorbers, and overtopping devices. These devices use different mechanical techniques such as a mechanical wave generation mechanism using a motor-driven system.

From the previous research carried out on wave energy conversion devices, it has been proven that these devices are capable of producing power for the coastal region where access to conventional electricity is limited. With the advancement of microcontroller and IoT technologies, these devices are now capable of being more efficient with the addition of real-time monitoring and control capabilities.

This project has been proposed based on the concepts of a wave-driven rotary system with IoT capabilities for better performance

1.2 Motivation

The main reason for undertaking this project is the increasing need for sustainable and eco-friendly forms of energy production. Conventional forms of energy such as coal and petroleum not only have limitations but also contribute to the degradation of the environment. It is therefore necessary to look for alternative forms of energy that are not only effective but also environmentally friendly.

Coastal areas are also endowed with wave energy, which is not yet exploited to its full potential. It is possible to generate electricity while at the same time not harming the environment. Moreover, these areas are also challenged by the lack of electricity due to their remoteness.

The other reason for undertaking this project is the integration of modern technologies such as IoT to enable real-time monitoring of the system. It is also possible to develop this form of energy to make it cheaper and applicable for other purposes.

1.3 Objectives

The research objectives form an essential part of this project as they clearly define the goals and expected outcomes of the study. The objectives are closely related to the problem of inefficient utilization of renewable energy resources, particularly in coastal regions where wave energy remains largely untapped. This project aims to explore a practical and sustainable solution to convert wave energy into electrical energy effectively.

The primary objective of this project is to design and develop a wave-driven rotary energy generation system that can efficiently harness ocean wave motion and convert it into usable electrical power. Another objective is to implement a mechanical energy conversion mechanism using a piston and turbine arrangement, ensuring effective transformation of kinetic energy into rotational motion.

Furthermore, the project aims to integrate an IoT-based monitoring system using a microcontroller such as Arduino Uno to enable real-time tracking of system parameters like voltage output and wave intensity. This helps in improving system efficiency and performance analysis.

Finally, the project seeks to develop a cost-effective, eco-friendly, and scalable energy solution that can be deployed in coastal and remote areas, thereby contributing to sustainable development and reducing dependence on conventional energy sources.

CHAPTER 2

METHODOLOGY

2.1 Applied Techniques and Tools

The proposed system is designed to employ various mechanical, electrical, and IoT-based methods to harness the wave energy and convert it into electrical energy. In this project, artificial wave motion is created by employing a motor-driven mechanism rather than depending on the motion of the waves in the ocean. For this purpose, a 1000 RPM side shaft motor is fixed to a rectangular plate to rotate continuously and agitate the water surface to create wave motion.

The artificial wave motion causes oscillations in the water, which eventually creates airflow within the system. The airflow is then directed through a duct and then towards a turbine called the Wells turbine, which is designed to rotate in one direction only and is capable of rotating regardless of the direction of airflow.

The mechanical motion is then transferred to a DC micro-generator to convert the mechanical motion into electrical motion. As the motion is irregular in nature, a bridge rectifier is used to convert the output into direct current and a capacitor to maintain the voltage at a stable level.

Moreover, an IoT-based system is also implemented within the system to monitor and analyze the parameters of the system. For this purpose, an Arduino Uno microcontroller is used along with a voltage sensor to measure the voltage output and transmit the data Matlab Platform.

2.2 Technical Specifications

The proposed system has been implemented by incorporating mechanical, electrical, and embedded components according to its requirements.

A side shaft DC motor rated at 1000 RPM is used for artificial wave generation.

The motor is mechanically coupled with a rectangular plate, and when it rotates, it causes disturbance on the water surface, resulting in wave motion.

The oscillatory motion generated by the system causes airflow, which is used by a Wells turbine.

The Wells turbine operates under bidirectional airflow, but its rotation is always unidirectional.

The rotation of the turbine shaft is mechanically coupled with a DC micro-generator, which converts mechanical energy into electrical energy.

The electrical energy is obtained from the DC micro-generator, and it is conditioned by using a bridge rectifier.

The output obtained from the rectifier is a DC signal, and a capacitor is used to minimize voltage ripple.

Arduino Uno is used as a microcontroller for data transmission.

A voltage sensor module is used and interfaced with the Arduino Uno microcontroller.

The Arduino Uno microcontroller has connectivity capabilities, and data is transmitted using its built- in computational and transfer capabilities.

The system has been implemented by using the Arduino IDE, and it is an efficient tool for programming and deploying IoT systems.

The data is transmitted to an IoT platform, and it is used for analysis.

The system has been implemented by using various components, and its requirements are met by incorporating them.

The output obtained from the DC micro-generator is used.The Arduino Uno microcontroller is used, and it has Bluetooth capabilities. The Arduino Uno microcontroller is used for data transmission, and it has been interfaced with a voltage sensor module.

The system has been implemented by using the Arduino IDE, and it is an efficient tool for programming and deploying IoT systems.

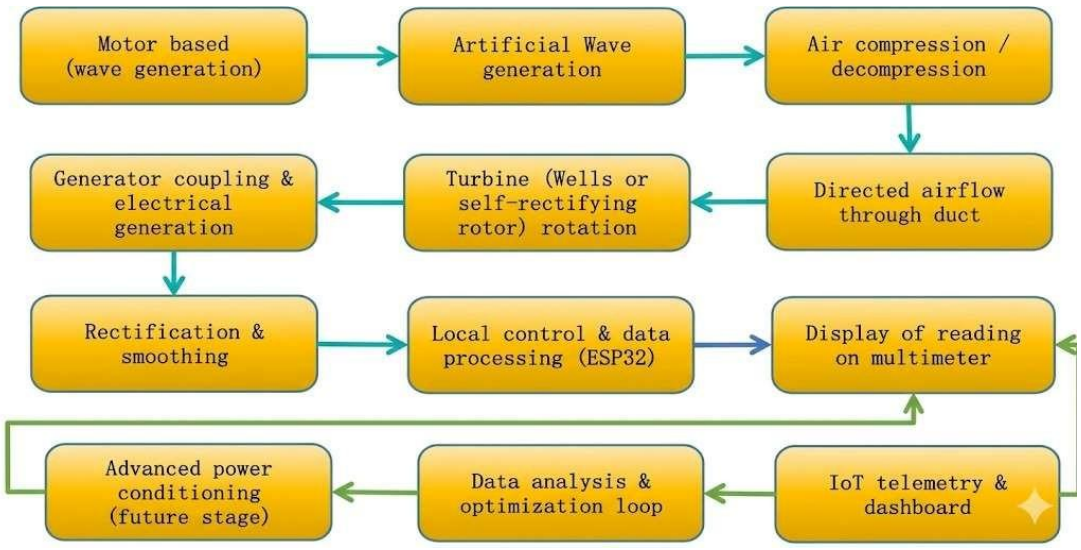
The data is transmitted to an IoT platform, and it is used for analysis the system.

2.3 System Architecture

The system architecture of the proposed model is composed of four main units. These include the wave generation unit, the energy conversion unit, the power conditioning unit, and the IoT monitoring unit. These units are interconnected to ensure the smooth operation of the system.

The wave generation unit of the proposed system is composed of a 1000 RPM side shaft motor and a rectangular plate. The rotating plate creates disturbances in the water, leading to the formation of waves.

The generated waves cause the oscillatory movement of the water, leading to the flow of air. This flow of air is then passed through a duct to the Wells turbine, constituting the energy conversion unit. This unit converts the kinetic energy of the airflow to mechanical energy. This mechanical energy is then transferred to a DC micro-generator for the generation of electrical energy.



Architecture Diagram of the system

The generated electrical output is then supplied to a power conditioning unit, which consists of a bridge rectifier and a capacitor. The purpose of this unit is the conversion of the generated DC output and the minimization of voltage fluctuations.

The IoT monitoring unit is implemented using an Arduino Uno microcontroller. A voltage sensor is connected with the system, which is used for the measurement of the output parameters. The Arduino Uno microcontroller then processes the data and sends it, using the cloud platform. The system thus achieves the efficient conversion of energy along with the analysis of the system.

2.4 Working Principle

The working principle of the proposed system is based on the idea of artificially generating wave energy and transforming it into electrical energy through a series of energy transformation processes.

The system first generates artificial waves by a motor-based mechanism. A 1000 RPM side shaft motor rotates a rectangular plate, causing the surface of the water to be disturbed and creating a wave-like effect.

The artificially generated waves cause oscillatory movements within the system, leading to airflow. The airflow is then passed through a duct and directed towards a Wells turbine.

The Wells turbine operates under oscillatory airflow conditions and transforms the kinetic energy of the airflow into mechanical energy. It rotates in a single direction regardless of the direction of the airflow.

The rotating action of the turbine shaft is connected to a DC micro-generator. It transforms the mechanical energy into electrical energy.

The transformed electrical energy is then passed through a bridge rectifier to obtain DC voltage. A capacitor is also connected to filter the voltage and reduce voltage fluctuations.

The system parameters are monitored by an Arduino Uno microcontroller through a voltage sensor. The data is then transmitted to the Matlab and IOT based platform.

2.5 Flow of Operation

The operation of the proposed system has a sequential flow from wave generation to the production of electric energy.

At first, the motor rotates the rectangular plate attached to it, causing disturbances in the water and resulting in wave generation.

This wave generation produces oscillatory motion, resulting in airflow within the system.

The airflow is then guided to a duct for the operation of the Wells turbine.

The Wells turbine converts the kinetic energy of the airflow into mechanical energy.

The turbine rotates continuously because of its ability to operate bidirectionally.

The rotating motion is then converted into electric energy with the help of a DC micro-generator.

The electric energy produced is then passed through a rectifier to produce DC voltage.

A capacitor is also used to smooth the voltage and reduce any voltage fluctuation.

The voltage sensor of the Arduino Uno microcontroller is then used to monitor the output voltage of the generator.

The operation of the proposed system provides a smooth flow of energy conversion along with efficient monitoring of the output voltage.

CHAPTER 3

EXPERIMENTATION AND TESTS

3.1 Experimental Setup

The experimental setup of the proposed system has been designed to prove the concept of artificially created wave energy being transformed into electrical energy. The setup has been arranged compactly and securely to ensure correct positioning of all components and accurate observation of system performance.

The setup has a water tank where waves are created artificially by a motor-driven mechanism. A 1000 RPM side shaft motor has been coupled with a rectangular plate that rotates continuously and creates a disturbance on the water surface. This creates a consistent and controlled wave form for experimentation purposes.

The artificially created waves cause oscillatory action within the system, which leads to airflow through a duct mechanism. This airflow is directed towards a Wells turbine that has been securely fixed to ensure system stability during experimentation. The turbine uses the kinetic energy of the airflow to convert it into mechanical energy.

The turbine shaft is connected to a DC micro-generator that uses mechanical energy to convert it into electrical energy. The generated electricity from the generator is passed through a bridge rectifier to convert it into direct current.

A capacitor is also connected to ensure smooth voltage output by reducing voltage fluctuation. A load is connected to indicate visually that power has been generated. An Arduino Uno microcontroller is connected with a voltage sensor to measure the generated voltage. Data is transmitted in real time for monitoring and analysis of the generated voltage and wave frequency.

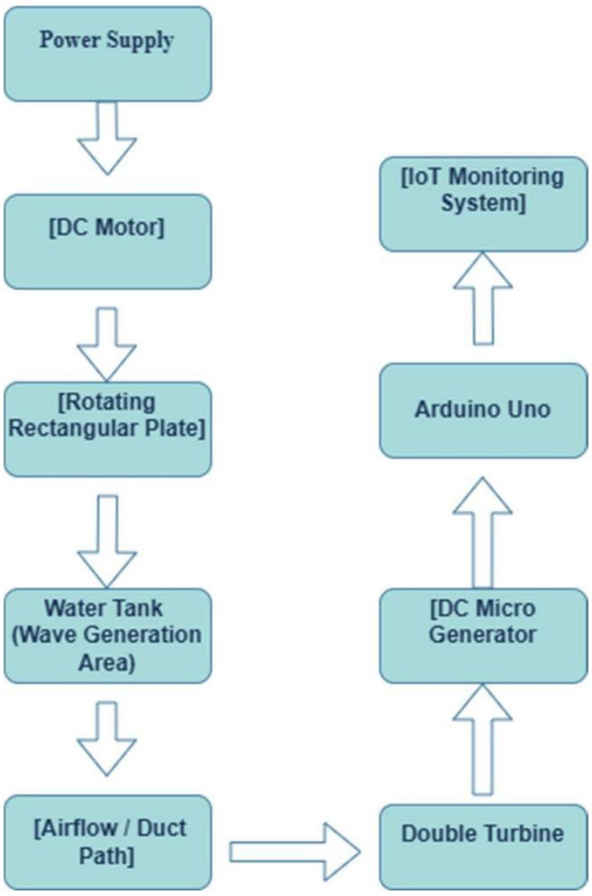


Fig. 3.1: Experimental Setup of the Wave-Driven Energy Generation System

3.2 Experimental Procedure

The experimental procedure is carried out in a systematic manner to evaluate the performance of the proposed system under controlled conditions. All components are initially assembled and checked to ensure proper connections and alignment.

The motor is powered on to initiate the rotation of the rectangular plate. As the plate rotates, it generates continuous wave motion in the water tank. The wave intensity depends on the speed of the motor and the size of the plate.

The generated waves produce oscillatory motion, which results in airflow within the system. This airflow is directed through a duct towards the Wells turbine. The turbine begins to rotate as a result of the airflow.

The rotational motion of the turbine shaft drives the DC micro-generator, which produces electrical output. The generated voltage is passed through a rectifier and capacitor circuit to obtain a stable DC output.

The output voltage is measured using a voltage sensor connected to the Arduino Uno microcontroller. The Arduino Uno processes the data and transmits it to an IoT platform for monitoring and analysis.

The experiment is repeated under different conditions by varying the motor speed or wave intensity. Observations are recorded for each trial to analyze system performance.

3.3 Prototype Testing

The experimental process is carried out in a systematic way to test the performance of the proposed system. All the components are initially assembled and connected properly.

The motor is turned on to start the rotation of the rectangular plate. As the plate rotates, it generates continuous wave motion in the water tank. The wave intensity depends upon the speed of the motor and the plate used in the system.

The generated wave motion causes oscillatory motion in the system. The oscillatory motion generates air flow in the system. The air flow generated by the oscillatory motion passes through a duct and reaches the Wells turbine. The rotation of the Wells turbine occurs due to the air flow generated by the oscillatory motion of the wave plate.

The rotation of the Wells turbine shaft causes the rotation of the DC micro-generator. The DC micro-generator generates electrical output. The generated voltage passes through a rectifier and capacitor circuit to produce stable DC voltage.

The generated voltage is measured by a voltage sensor connected to the Arduino Uno microcontroller. The Arduino Uno microcontroller processes the data and sends it to the IoT platform.

The experiment is repeated under different conditions by varying the speed of the motor. The results are obtained by observing the system. Electrical energy, while also providing reliable monitoring through IoT integration.

CHAPTER 4

CHALLLEGES, CONSTRAINTS AND STANDARDS

4.1 Challenges and Remedies

During the design, implementation, and testing of the proposed system, a number of technical and practical difficulties were encountered. These difficulties were thoroughly studied and appropriate remedies were applied to ensure smooth operation of the proposed system.

One of the main difficulties was to ensure uniform wave motion using the proposed mechanism of a motor. The irregularity in the speed of the motor and the irregular rotation of the rectangular plate caused difficulties in wave formation. This difficulty was overcome by maintaining a constant voltage for the motor and balancing the plate properly.

Another difficulty was to ensure efficient conversion of wave-induced air flow into mechanical energy. The unsteadiness of the air flow caused by the oscillatory motion of the waves was another problem that needed to be overcome. The unsteadiness of the air flow resulted in irregular rotation of the turbine. This difficulty was overcome by enhancing the design of the duct to ensure effective air flow.

Voltage instability was another difficulty that was encountered during the implementation of the proposed system. The voltage was fluctuating in nature and could not be used directly for practical purposes. This difficulty was overcome by using a bridge rectifier and capacitor.

The mechanical alignment of the turbine, generator, and coupling was another difficulty that was encountered during the implementation of the proposed system. The improper alignment of these components resulted in energy losses. Proper alignment of these components was ensured to overcome this difficulty.

Environmental factors such as water splashing also caused difficulties in implementing the proposed system. Proper arrangements were made to overcome these difficulties.

4.2 Design Constraints

The design of the proposed system is affected by various constraints. These constraints have been considered while developing the system to provide a balanced and viable design.

The first constraint is the limitation of resources and budget. The proposed system is designed to utilize low-cost and easily available resources. This is the major constraint in the development of the proposed system. This affects the performance of the proposed system.

The space constraint is another constraint in the development of the proposed system. The proposed system is designed for small-scale implementation. This affects the performance of the proposed system.

The proposed system is designed to operate with artificial waves. This is another constraint in the development of the proposed system. This affects the performance of the proposed system.

The power generated by the motor and generator is another constraint in the development of the proposed system. The proposed system is designed for small-scale implementation. This affects the performance of the proposed system.

The material and mechanical design of the proposed system also affect the performance of the proposed system.

4.3 Alternatives and Trade-offs

During the design stage, a number of alternatives were considered to enhance the performance and feasibility of the system. However, some trade-offs had to be made to ensure balance.

One of the alternatives was the use of natural ocean waves. Although natural waves are associated with higher energy potential, they are difficult to control during a laboratory experiment. Therefore, a motor-driven system was considered more viable.

Different designs of turbines were considered for use in the system. Although a highly efficient turbine would increase system performance, it would also increase the complexity of the system. Therefore, the use of a Wells turbine was considered viable for its simplicity and ability to operate under bidirectional airflow.

Another trade-off was the use of a system that was either efficient or cost-effective. Although a highly efficient system would increase the performance of the system, it would also increase the overall cost of the project. Therefore, a cost-effective approach was considered viable.

The level of complexity was also considered during the design stage. Although a highly complex system would increase the performance of the system, it would also increase the complexity of the system. Therefore, a simpler system was considered viable.

These trade-offs ensured that the system was functional, cost-effective, and viable.

4.4 Standards

The proposed system is designed and implemented considering various engineering and communication standards to ensure reliability and safety.

The proposed IoT communication system is based on IEEE 802.11 Wi-Fi standards for wireless data transfer between the Arduino Uno microcontroller and the cloud platform.

The safety of the electrical components is ensured by following standard low voltage safety practices to prevent any electrical hazards.

The mechanical aspects of the proposed system are designed considering various engineering standards to ensure proper component orientation and stability to avoid any mechanical issues.

This also helps to prevent any loss of energy and to make the system highly efficient.

The programming of the embedded system is done considering standard programming practices and the Arduino development environment.

Furthermore, general environment and safety considerations are also taken into account to ensure safe handling and operation of the system.

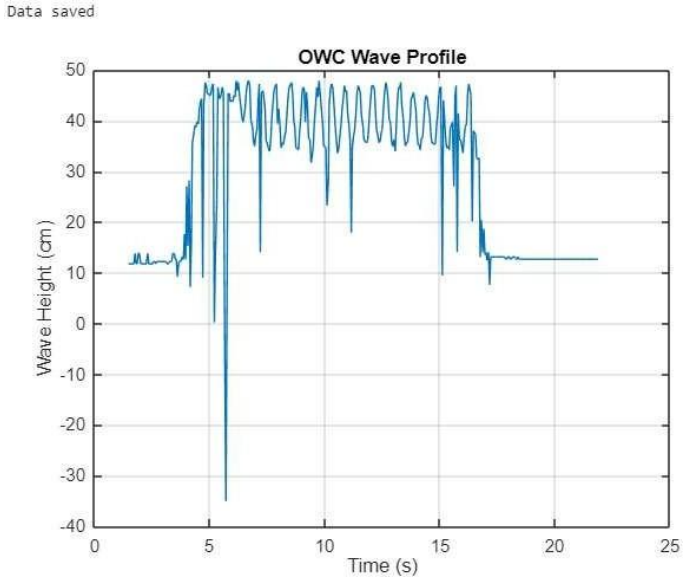
CHAPTER 5

RESULT ANALYSIS AND DISCUSSION

5.1 Results Obtained

The proposed system is tested for various operating conditions to assess its performance. The voltage generated by the generator is noted for various levels of wave intensity produced by the motor-driven mechanism.

From the observations, it is clear that as the intensity of wave generation is increased, the speed of the turbine also increases to achieve higher voltage output. The system is capable of generating voltage for all the operating conditions..



5.1.i The experimental results obtained are summarized in Table

TABLE TYPE STYLES

	A	B	C
	wavedata		
	Time_s	Distance_cm	WaveHeig...
	Number	Number	Number
1	Time_s	Distance_cm	WaveHeigh...
2	1.5219	13.3800	36.6200
3	1.5399	9.9100	40.0900
4	1.6035	5.3000	44.7000
5	1.6499	5.2000	44.8000
6	1.7032	2.0900	47.9100
7	1.7592	2.7400	47.2600
8	1.8065	4.0300	45.9700
9	1.8696	6.2700	43.7300

5.1.ii Obtained Wave data



5.1.iii Obtained Current Reading at 20ma on mutimeter

Time (sec)	Current (ma)
0	0.0
1	-3.4
2	-3.6
3	0
4	-4.28
5	-4.28
6	0
7	-4.2
8	0

5.1.iv Obtained mltimeter Output in Table format

5.2 Analysis and Discussion

The results obtained from the experimental analysis of the proposed system show that the proposed model is effective in converting artificially generated wave energy into electric energy. There exists a strong relationship between the intensity of the waves and the electric energy produced by the proposed system.

As the speed of the motor increases, the intensity of the waves also increases proportionally. This results in more oscillatory motion and increased airflow within the system. The increased level of airflow results in increased speed of the Wells turbine, causing more mechanical energy to be converted into electric energy.

The performance of the Wells turbine was also found to be stable for oscillatory airflow conditions. The design of the Wells turbine ensured that it rotated even when the direction of the airflow changed. This is a critical aspect of wave-based systems.

The implementation of the rectifier and capacitor circuit was also effective in enhancing the quality of the output voltage produced by the proposed system. The rectifier ensured that the output voltage was converted to direct current, while the capacitor ensured that the voltage was smooth.

The IoT monitoring system also functioned efficiently for the proposed system. The Arduino Uno microcontroller was effective in measuring the voltage and transmitting it in real time.

However, some limitations were also observed while analyzing the proposed system. One of the limitations of the proposed system was the presence of energy loss at different stages of conversion. In addition, the use of artificially generated waves may have a negative impact on the efficiency of the proposed system when it is implemented for real-world scenarios.

5.3 Project Demonstration

The project was successfully demonstrated, validating the working principle and performance of the proposed system. The demonstration was carried out in complete laboratory settings.

The motor-driven mechanism was first activated, causing the rotation of the rectangular plate, thus generating wave motion in the water tank. The generated waves produced oscillatory motion, causing the airflow in the system.

The airflow was effectively directed to the Wells turbine, causing it to rotate continuously. The rotation of the shaft was transferred to the DC micro-generator, causing the generation of electrical energy. The voltage was observed and verified using the connected load, represented by the multimeter reading, which successfully glowed, indicating the generation of electrical power.

The generated electrical output was then passed through the rectifier and capacitor circuit, thus stabilizing the voltage level. The improvement in voltage stability was clearly observed during the demonstration.

The IoT-based monitoring system was also successfully demonstrated. The Arduino Uno microcontroller was able to collect voltage data from the system, thus transmitting it to the cloud-based platform.

The demonstration has clearly proved the working principle and performance of the system, indicating its capability to convert wave energy into electrical energy. The system also has the potential for small-scale development.

CHAPTER 6
CONCLUSIVE REMARKS

6.1 Project Planning, Progress and Management

The development of the proposed wave-driven energy generation system was done in a systematic and well-structured manner through proper planning, execution, and evaluation of the project. The development of the proposed wave-driven energy generation system was done through various phases to ensure smooth workflow and proper utilization of time and resources.

In the beginning, the problem statement and objectives of the project were clearly defined, focusing on the need for renewable energy solutions and the potential of wave energy as a sustainable source of power. The required methodology for the development of the proposed wave-driven energy generation system was identified and properly planned.

During the design phase of the development of the proposed wave-driven energy generation system, various methodologies were identified and discussed for the development of the wave-driven energy generation system. It was decided to develop a motor-driven wave generation mechanism to simulate wave conditions in a controlled environment. The required components of the wave-driven energy generation system, such as the motor, turbine, generator, Arduino Uno microcontroller, and sensors, were properly selected.

During the implementation phase of the development of the proposed wave-driven energy generation system, the required system components were assembled properly to ensure smooth integration and functionality of the system. The system was properly tested at various stages to ensure smooth functionality and to identify any issues in the system. The required modifications were done to improve the functionality of the system.

The development of the proposed wave-driven energy generation system was properly monitored and completed within the scheduled time to achieve the objectives of the project through proper time management and coordination of the tasks.

6.2 Conclusion

The proposed wave-driven rotary energy generation system is successful in demonstrating the feasibility of converting wave energy into electrical energy through a motor-driven mechanism.

The experimental results of the proposed system prove its efficiency in generating electricity from wave energy. The efficiency of the Wells turbine and DC generator is also proved to be consistent in the experiment. Moreover, the power conditioning circuit is also proved to be effective in stabilizing the voltage.

The integration of the Arduino Uno microcontroller into the proposed system is also proved to be effective in enhancing the functionality of the system. The microcontroller is effective in providing real-time monitoring and data transmission.

The proposed system is also successful in demonstrating the potential of wave energy as a renewable source of energy. The proposed system is also successful in demonstrating the potential of wave energy as a source of electricity.

In conclusion, the proposed system is successful in achieving its objectives and is also effective in advancing the field of sustainable energy development.

From the experiment and analysis of the proposed system, the following conclusions are made:

The proposed wave-driven rotary energy generation system is successful in demonstrating the efficiency of wave energy as a source of electricity.

The proposed system is also successful in demonstrating the potential of wave energy as a renewable source of energy.

The proposed system is successful in achieving its objectives.

The proposed system is effective in advancing the field of sustainable energy development.

6.3 Further Plan of Action / Future Work

The proposed system is a good platform for future research and development in the field of wave energy conversion. Several changes and improvements can be incorporated to improve the efficiency of the system.

The first major change is the implementation of the natural ocean waves rather than artificial ones. This would improve the efficiency of the system as the system would be able to work in the real world. In addition to this, the implementation of more efficient and advanced turbines can also be carried out.

The proposed system can also be improved by increasing the size of the components and the mechanical structure to increase the power output. In addition to this, the efficiency of the airflow and the turbines can also be optimized to improve the performance of the system.

The proposed system can also be improved by the implementation of advanced IoT features. In addition to this, the proposed system can also be combined with other renewable energy resources such as solar and wind to form a hybrid system. This would ensure the availability of the required power.

The proposed project is a good platform for future innovation and implementation of the proposed system.

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Appendix A: Gantt Chart

	Dec.15-25	Dec.28-Jan 15	Jan.25-Feb. 1	Feb.10-Feb. 23	Mar. 20
Background Studies/Literature Survey					
Problem Identification& Objective Setting					
System Design & Architecture Planning					
Component Selection (Motor, Arduino Uno, etc.)					
Development of Wave Generation Mechanism					
Circuit Setup & Arduino Uno Integration					
prototype Assembly					
Testing & Data Collection					
Analysis & Optimization					
Report Writing					
Final Presentation Preparation					

Appendix B: Project Summary

Project Title	Wave-Driven Rotary Energy Generation System for Coastal Communities
Team Members	Ankit Kumar, Vibhore Krishna, Suprit Mangaraj, Mukulika Bhattacharyya
Supervisors	**Prof. Manoj Kumar Parida**

Semester / Year	**VI / III year**
Project Abstract	The growing demand for the use of sustainable and renewable forms of energy has led to the exploration of alternative methods for generating power. The current project proposes the design of a Wave-Driven Rotary Energy Generation System, which can be utilized for the efficient conversion of wave energy into electrical power. The system works on the principle of utilizing the kinetic energy produced due to the motion of the waves, which is then converted into mechanical form using the motor-driven wave generation system. The mechanical form of the wave's motion is then utilized for the compression of air, which in turn drives the turbine connected to the micro-generator, thus generating electrical power. For the efficient generation of electrical power, the generated power is regulated using rectifiers and capacitors, thus helping in the stabilization of voltage. In addition, the IoT system has also been incorporated using the Arduino Uno microcontroller, thus helping in the efficient tracking of voltage, wave intensity, and the overall performance of the system. The proposed system is efficient, cost-effective, eco-friendly, and can be implemented in coastal areas where conventional sources of energy are scarce. The system also provides the advantage of further development for large-scale implementation. The current project proposes the efficient conversion of wave energy into electrical power using the hybrid approach. The current system contributes to the development of renewable energy, thus helping in the achievement of the concept of sustainable development through the efficient utilization of the natural resources present in the ocean.

List **codes** and **standards** that significantly affect your project.	• IEEE 802.11 (Wi-Fi) standard used for wireless communication via Arduino Uno • Low-voltage electrical safety standards for DC generator and motor operation • Embedded system programming standards using Arduino IDE •Standard IoT communication protocols for data transmission and monitoring
List at least two significant realistic design constraints that are applied to your project.	• Limited ability of artificially generated waves to replicate real ocean conditions • Use of low-cost components (motor, turbine, generator) reduces overall efficiency • Small-scale setup restricts power output and scalability • Dependence on stable power supply for motor operation.

Briefly explain two significant trade-offs considered in your design, including options considered and the solution chosen	• Trade-off between cost and efficiency: High-efficiency turbines and generators could improve performance, but low-cost components were chosen to keep the system economical and easy to implement. • Trade-off between system complexity and simplicity: A more complex design could provide higher accuracy and control, but a simpler design was selected to ensure ease of construction, operation, and maintenance.
Describe the computing aspects, if any, of your project. Specifically identifying hardware software trade-offs, interfaces, and/or interactions	• The Arduino Uno microcontroller is used for data acquisition and processing, where it reads voltage data from the sensor and controls system monitoring. • The system uses Wi-Fi communication (IEEE 802.11) to transmit real-time data to an IoT platform, enabling interaction between hardware and software components.
Culminating Knowledge and lifelong learning experience	For this project, knowledge from Microprocessors and Microcontrollers Digital Signal Processing Embedded Systems Design Wireless and Mobile Communication Internet of Things (IoT) Renewable Energy Systems and Wave Energy Concepts subjects has been used.